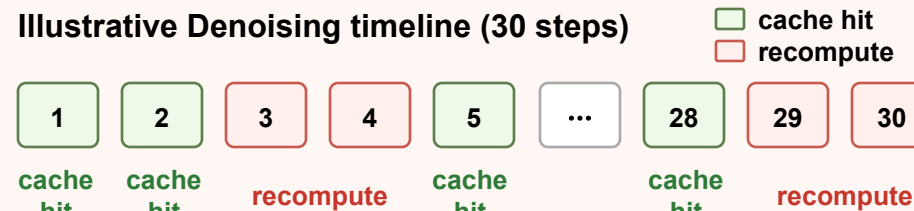


Challenges

1 Decision-Domain Mismatch in Cache+Quant

Illustrative Denoising timeline (30 steps)

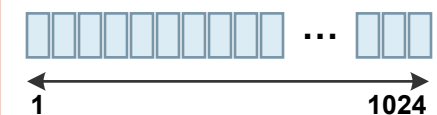


FP indicator $\neq$ INT8 execution relevance



2 Matrix-Axis Mismatch in Factorized Attention (64 Frames, 512×512 Resolution example)

Spatial: 38 (19x2) contexts × 1024-token rows



Temporal: 2048 (1024x2) contexts × 19-token rows



54× more contexts

QKT: $[1, d] \times [d, L] \rightarrow$ irregularity on output axis

PV: $[1, L] \times [L, d] \rightarrow$ irregularity on reduction axis

Proposed Co-Design

1 QAC: Quantization-Aligned Caching

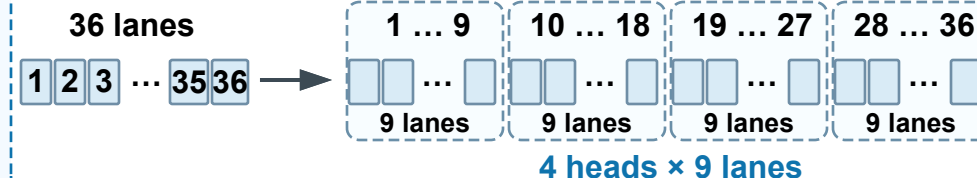
Quantized feature change estimation

Calibrated predictor

Cache hit / Recompute decision

2 DARE: Dimension-Adaptive Reconfigurable Engine

① QK output-axis adaptation



② PV reduction-axis adaptation

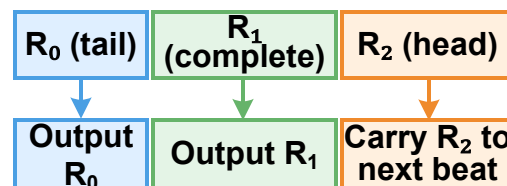
Baseline: padding for short rows (e.g., $T = 19$)

19 valid products 17 padding (idle)



One dot-product result

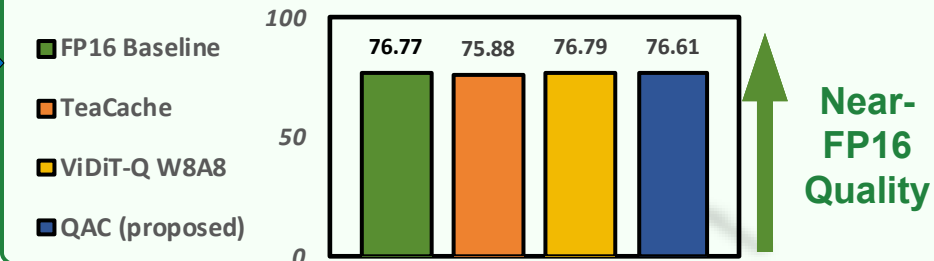
Proposed: boundary-aware segmented reduction



Benefits / Results

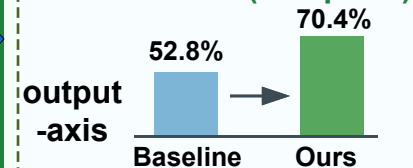
1 Algorithm Quality

VBenCh (higher is better)

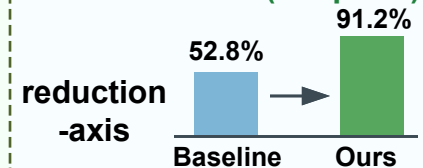


2 Hardware Efficiency

QK utilization (Temporal)



PV utilization (temporal)



3 System-Level Gains

Throughput:	10.96× vs A100; 1.06× vs ViDA
Energy Efficiency:	374.7× vs A100
Area Efficiency:	24.07× vs ViDA